



DATA PREPROCESSING AND EDA LAB PROJECT REPORT

Road Traffic Accident (RTA) Severity Analysis & Prediction

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1. INTRODUCTION

Road traffic accidents are one of the most pressing public health and safety concerns of our time. According to the World Health Organization, road traffic injuries claim approximately 1.35 million lives each year, with a disproportionate burden falling on low- and middle-income countries. Ethiopia, like many developing nations, struggles with a high incidence of road traffic accidents driven by a combination of infrastructure challenges, rapid urbanization, and varying levels of driver awareness.

This project attempts to carry out an analysis of a real-life dataset on road traffic accidents collected in Ethiopia, comprising 12,316 entries for accident cases.

The dataset contains 32 attributes ranging from driver characteristics, car information, weather condition, road environment, and time of accident. The outcome variable, Accident_severity, assigns each accident as having Slight Injury, Serious Injury, or Fatal Injury.

It is aimed at conducting full preprocessing and EDA, deriving useful insights, and implementing machine learning techniques like Logistic Regression, Decision Trees, and Random Forests for predicting accident severity.

2. DATA ACQUISITION

The dataset was sourced from Kaggle, originally compiled by Ethiopian police and traffic management authorities. It reflects genuine on-ground accident records from multiple regions.

The file was provided in CSV (Comma-Separated Values) format and loaded into a Jupyter Notebook environment using pandas' read_csv() function for all subsequent preprocessing and analysis.

3. DATASET DESCRIPTION

The dataset is rich and multidimensional, capturing accident scenarios from multiple angles. With 32 features covering driver demographics, vehicle information, road and environmental conditions, and casualty details, it provides a holistic view of the circumstances surrounding each accident. The target variable is Accident_severity, which has three classes: Slight Injury (the most common, approximately 84.6%), Serious Injury, and Fatal Injury (approximately 1.3%). This class imbalance is an important consideration during analysis and model building.

Two numerical columns — Number_of_vehicles_involved and Number_of_casualties — were the primary quantitative features, while the remaining 30 columns were categorical in nature. This heavy categorical profile makes encoding a critical step before any machine learning pipeline can be applied.

Column Name	Description	Data Type
Time	Exact time of accident occurrence (HH:MM:SS)	String → Hour (Int)
Day_of_week	Day the accident occurred	Categorical
Age_band_of_driver	Age group of the driver	Categorical
Sex_of_driver	Gender of the driver	Categorical
Educational_level	Driver's educational background	Categorical
Vehicle_driver_relation	Owner/employed/other relation to vehicle	Categorical
Driving_experience	Years of driving experience	Categorical
Type_of_vehicle	Type of vehicle involved	Categorical
Owner_of_vehicle	Who owns the vehicle	Categorical
Service_year_of_vehicle	Age/service years of the vehicle	Categorical
Defect_of_vehicle	Any reported vehicle defect	Categorical
Area_accident_occured	Urban/rural area of accident	Categorical
Lanes_or_Medians	Road lane type	Categorical
Road_alignment	Road alignment description	Categorical
Types_of_Junction	Junction type at accident spot	Categorical
Road_surface_type	Type of road surface	Categorical
Road_surface_conditions	Condition of road at time of accident	Categorical
Light_conditions	Lighting at time of accident	Categorical
Weather_conditions	Weather at the time of accident	Categorical
Type_of_collision	Collision type (rear-end, head-on, etc.)	Categorical
Number_of_vehicles_involved	Count of vehicles in accident	Numerical
Number_of_casualties	Total casualties in the accident	Numerical
Vehicle_movement	Vehicle movement at time of accident	Categorical
Casualty_class	Class/type of casualty	Categorical
Sex_of_casualty	Gender of casualty	Categorical
Age_band_of_casualty	Age group of casualty	Categorical
Casualty_severity	Severity of casualty	Categorical
Work_of_casualty	Occupation of casualty	Categorical
Fitness_of_casualty	Medical fitness of casualty	Categorical

Pedestrian_movement	Pedestrian movement if involved	Categorical
Cause_of_accident	Primary cause of the accident	Categorical
Accident_severity	TARGET — Slight/Serious/Fatal Injury	Categorical (Target)

4. DATA PREPROCESSING AND CLEANING

4.1 Initial Assessment

Using `df.info()`, `df.describe()`, and `df.shape`, the dataset structure was examined. Defect_of_vehicle (35.9% missing) and Service_year_of_vehicle (31.9% missing) had the highest missing rates. No duplicate rows were found initially.

4.2 Handling Missing Values

Mode imputation was applied to all categorical columns, replacing missing values with the most frequent category. For numerical columns, median imputation was used as it is more robust to outliers than the mean.

4.3 Duplicate Removal & Inconsistent Values

Duplicate rows were identified and removed using `drop_duplicates()`. Inconsistent text casing in Lanes_or_Medians and Types_of_Junction ('other' vs 'Other') was standardised to 'Other' across both columns.

4.4 Feature Engineering — Time Column

The Time column (string format HH:MM:SS) was converted into two new features: Hour (integer, 0–23) and Time_of_Day (Morning / Afternoon / Evening / Night). The original Time column was then dropped, reducing noise while adding interpretable temporal features.

5. DATA ENCODING AND SCALING

5.1 Label Encoding

Label Encoding was applied to all categorical columns to produce the `df_encoded` version. Each unique category is assigned an integer value — suitable for tree-based models like Decision Tree and Random Forest that do not assume ordinality.

5.2 One-Hot Encoding

One-Hot Encoding was applied using `pd.get_dummies()`, creating a binary column for each unique category. This avoids false ordinal relationships and is better suited for linear models like Logistic Regression.

5.3 Standardization and Normalization

`StandardScaler` was applied to numerical columns, transforming values to mean=0 and std=1. Additionally, `MinMaxScaler` was used to normalize values to the [0,1] range which is necessary for distance-based algorithms like KNN.

6. EXPLORATORY DATA ANALYSIS AND VISUALS

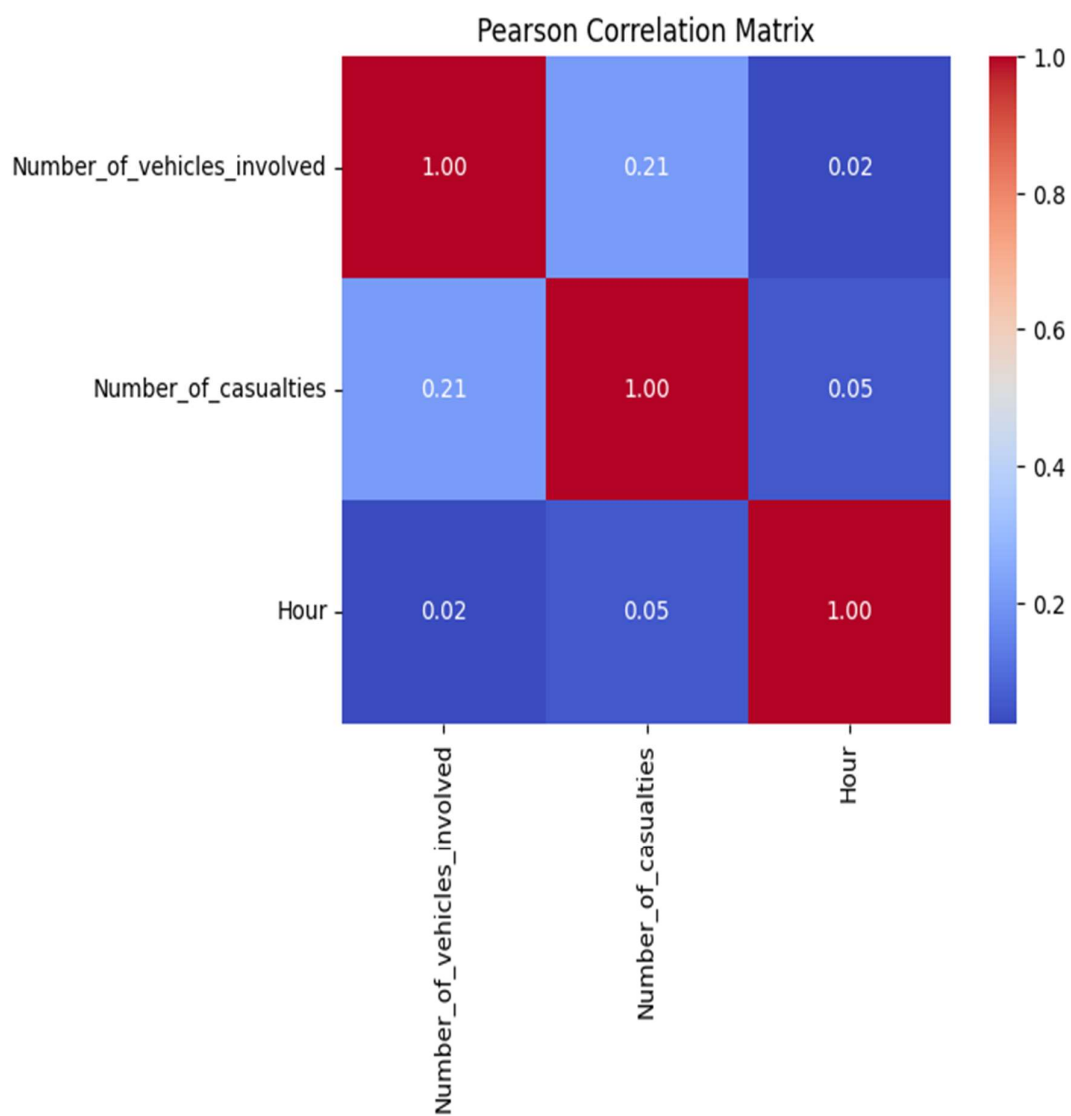
Exploratory Data Analysis (EDA) is the process of systematically examining a dataset using statistical summaries and visualisations to discover patterns, detect anomalies, and build an intuitive understanding of the data before applying any formal modelling. In this project, EDA was carried out across four levels of increasing complexity: grouping and ranking analysis, correlation analysis, univariate analysis, bivariate analysis, and multivariate analysis.

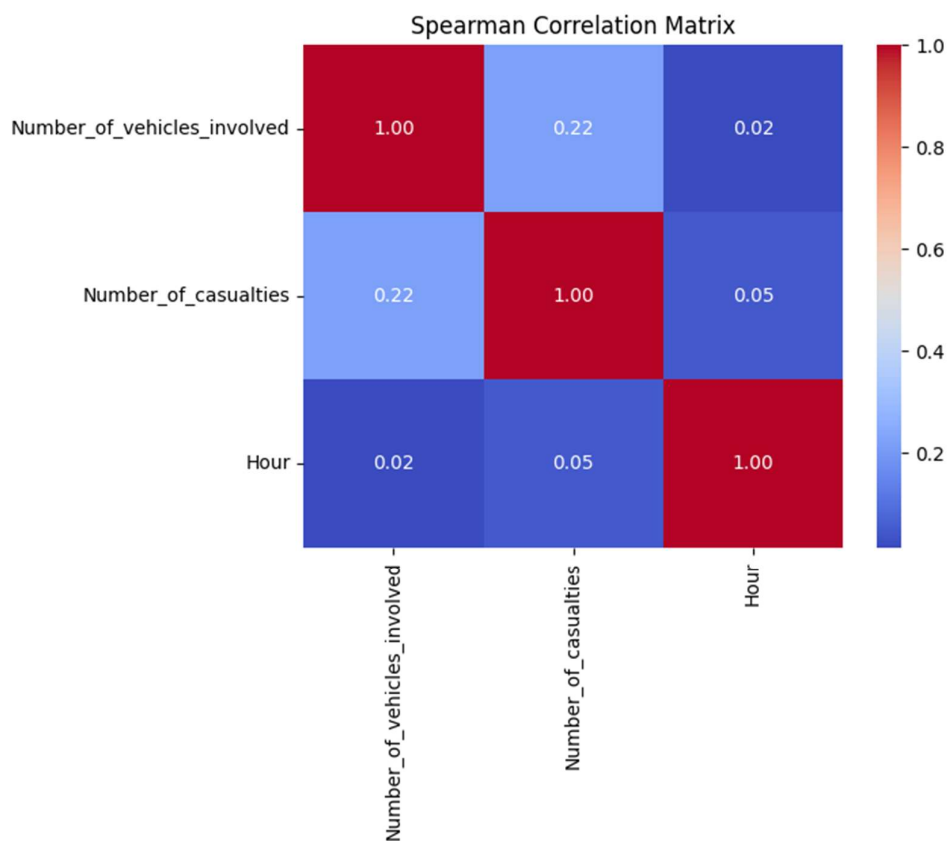
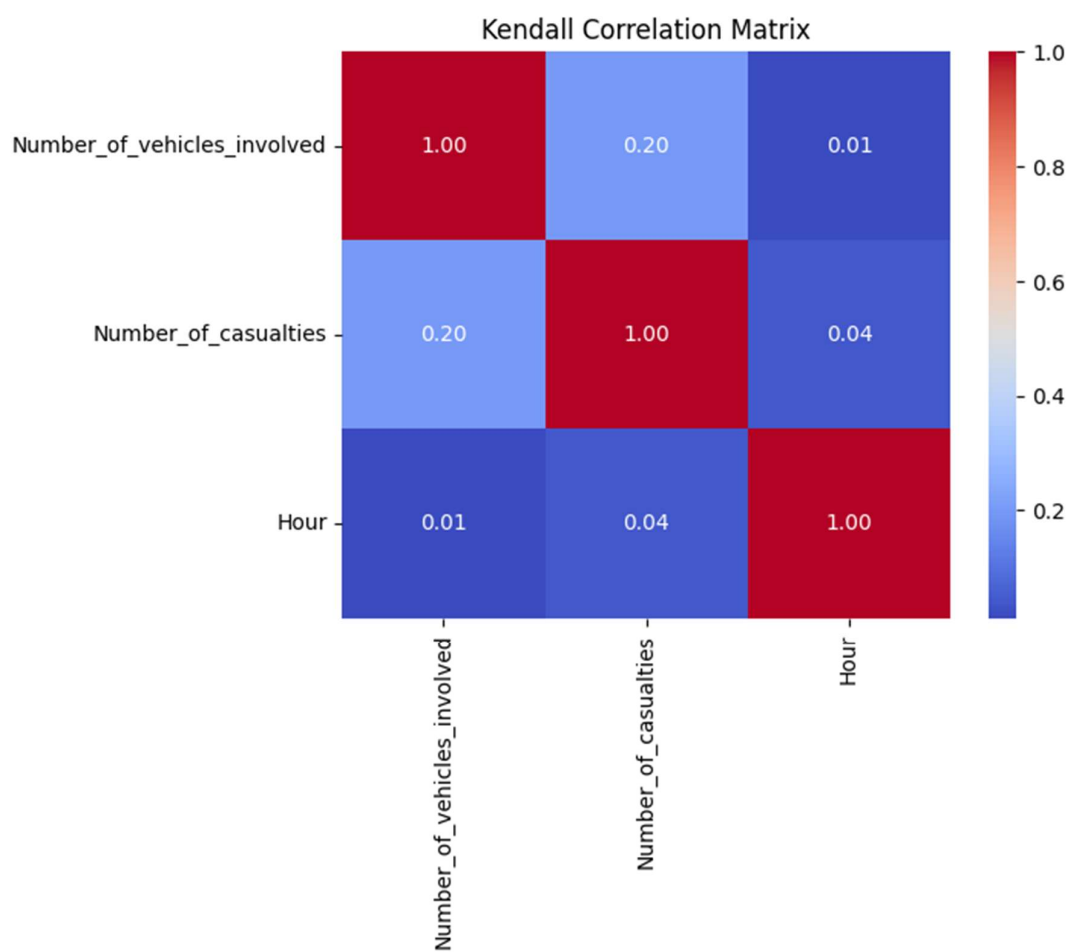
6.1 Grouping, Sorting, and Ranking

The first set of analyses involved grouping the data by key dimensions. The average number of casualties was computed for each level of accident severity using `groupby()`, revealing that fatal accidents unsurprisingly involve more casualties on average. The top 10 most frequent causes of accidents were identified using `value_counts()`, with 'no distancing' and 'overtaking' emerging as the leading causes. Finally, accidents were ranked by day of the week, with Friday consistently recording the highest accident count.

6.2 Correlation Analysis

Correlation analysis was performed using three methods: Pearson, Kendall, and Spearman. Pearson correlation measures linear relationships between numerical variables. Kendall and Spearman correlations are rank-based and are more robust to non-linear relationships and outliers. All three correlation matrices were visualised using seaborn heatmaps. The analysis consistently showed that the `Number_of_casualties` has the strongest positive correlation with `Accident_severity` — a logical and expected finding. Most other features showed weak or negligible pairwise correlations with each other, suggesting limited multicollinearity.

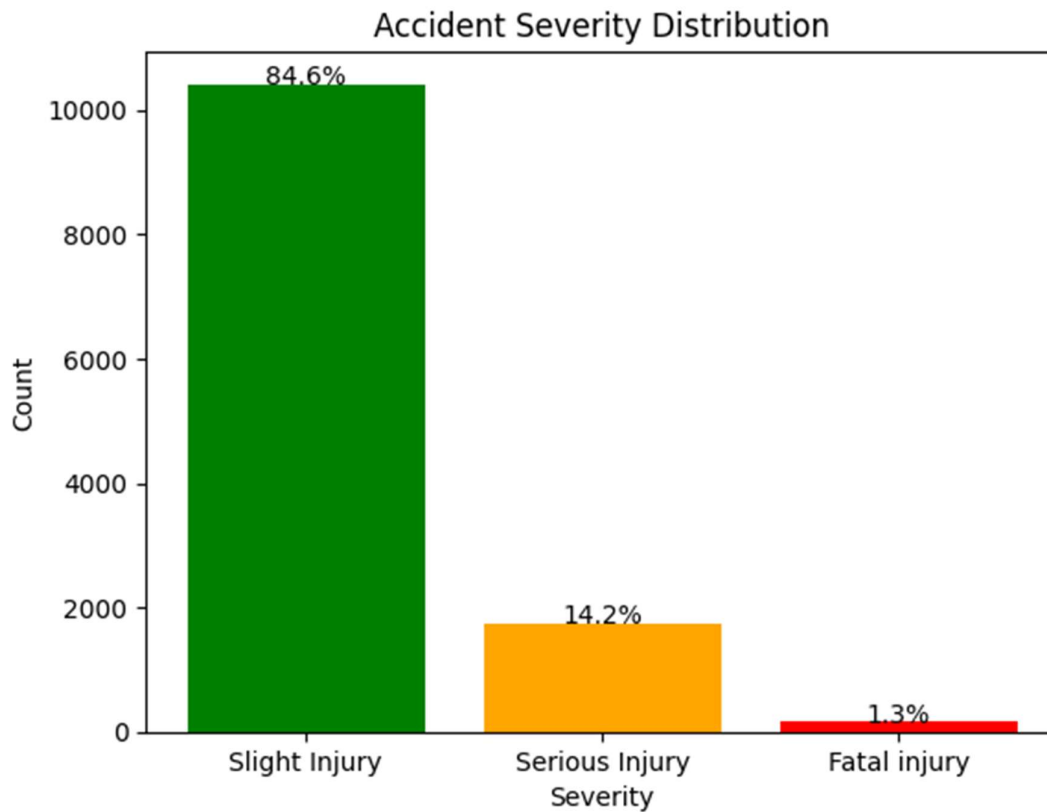




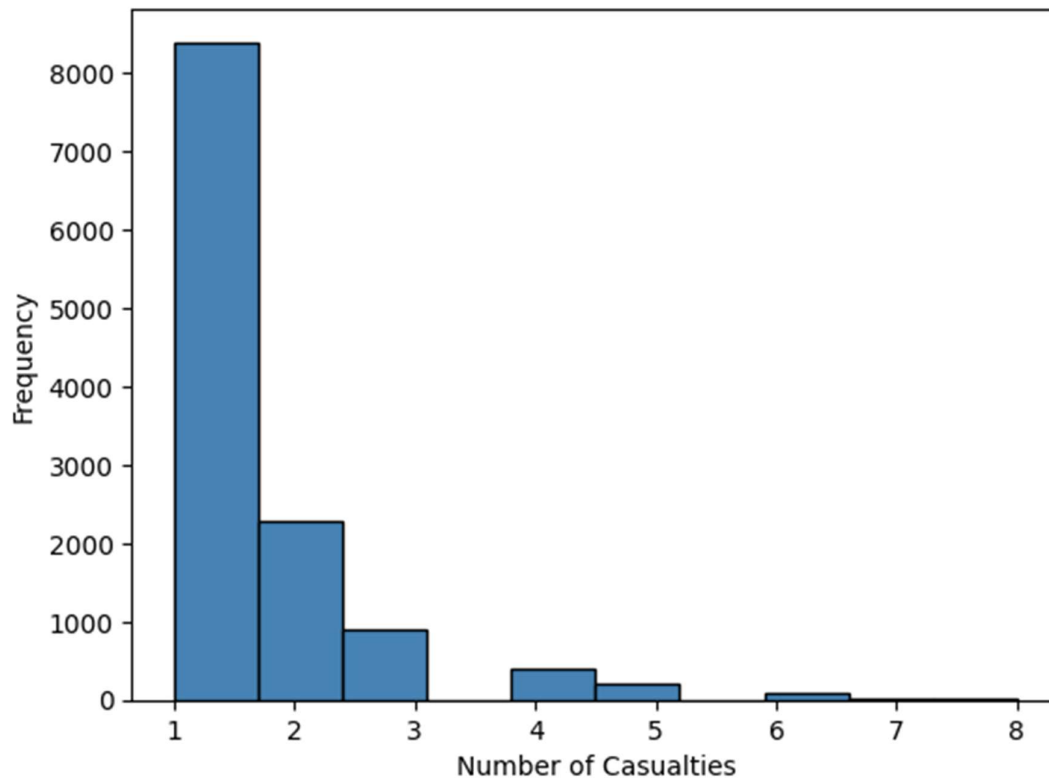
6.3 Univariate Analysis

Univariate analysis examines each variable in isolation to understand its distribution and characteristics. The distribution of the target variable, Accident_severity, confirmed a heavily imbalanced dataset: approximately 84.6% of accidents resulted in Slight Injury, around 14.1% in Serious Injury, and only about 1.3% were Fatal. This imbalance is a real-world reflection of road accident data and was noted as an important consideration for model training.

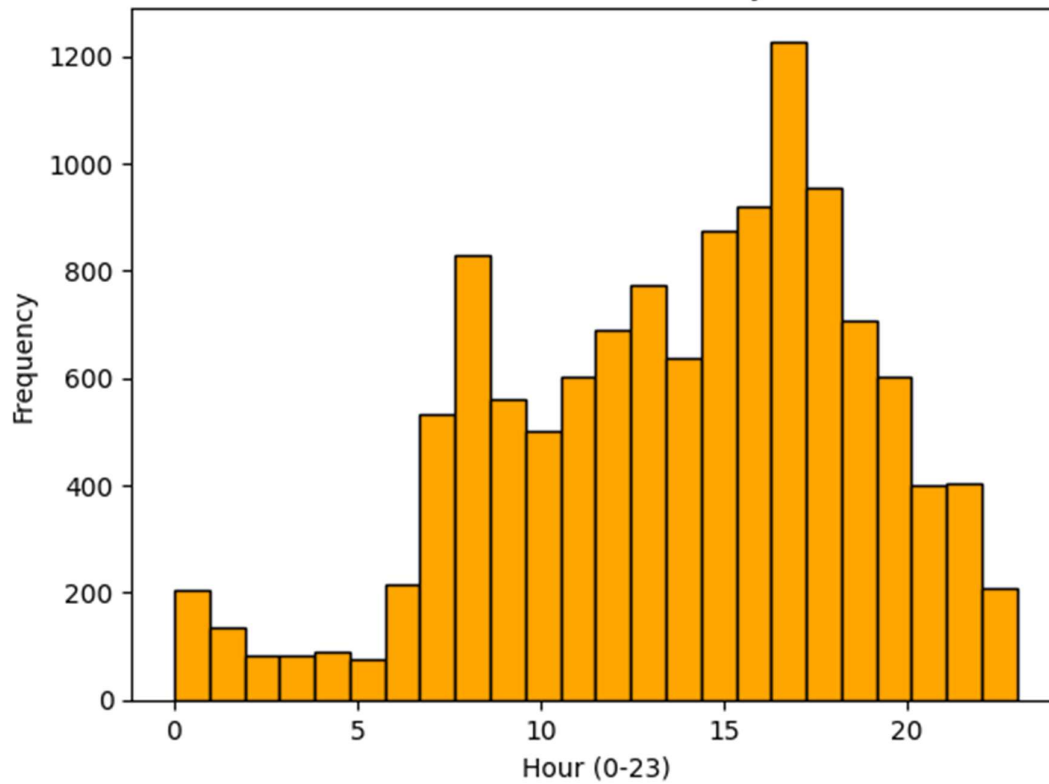
A histogram of Number_of_casualties showed a right-skewed distribution, with most accidents involving just one or two casualties — consistent with the dominance of Slight Injury cases. The distribution of accidents by hour of day revealed a clear peak in the 17–19 hour window (5pm–7pm), characteristic of evening rush hour traffic. The driver age band analysis showed that the 18–30 age group was involved in the highest number of accidents, followed by the 31–50 group. Among days of the week, Friday had the most accident occurrences, with the pattern suggesting end-of-week fatigue or increased social travel.



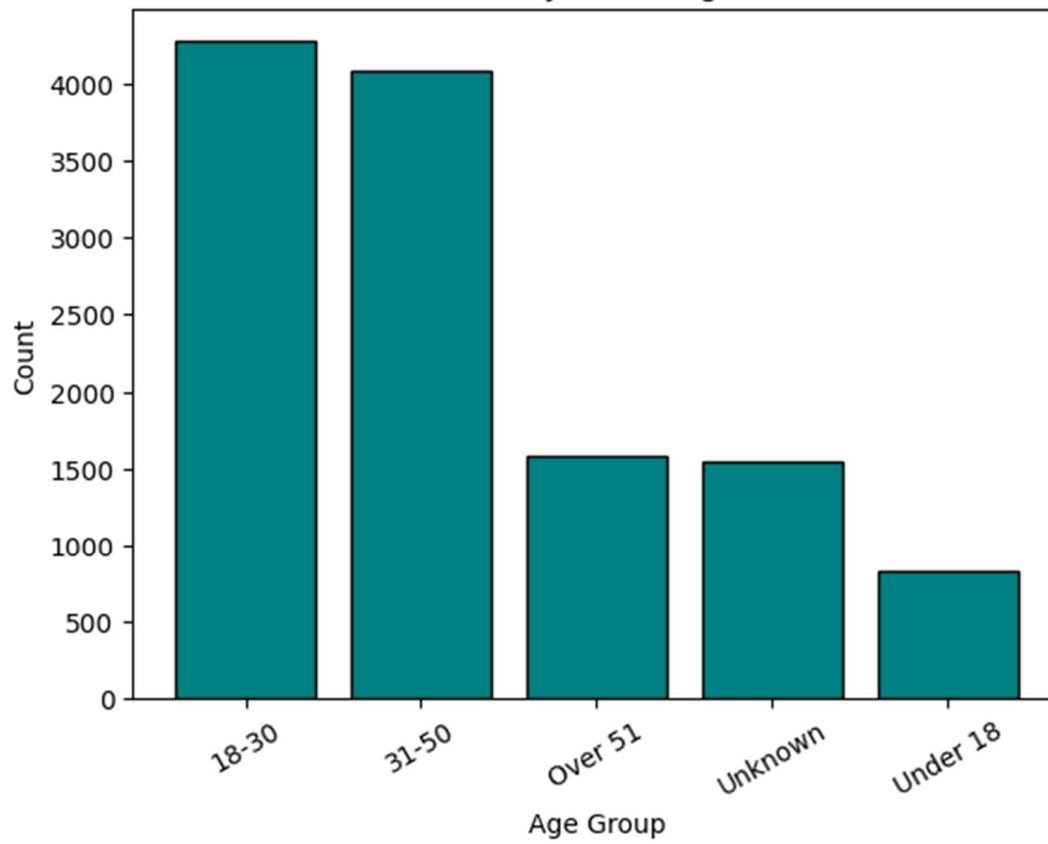
Distribution of Number of Casualties



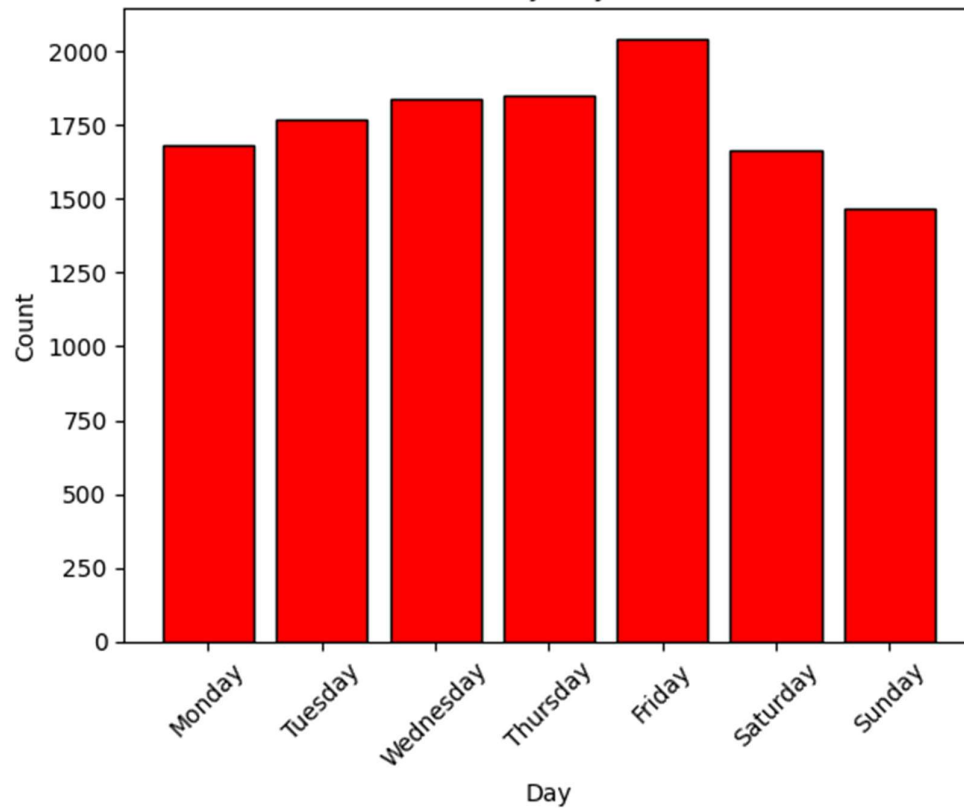
Distribution of Accidents by Hour

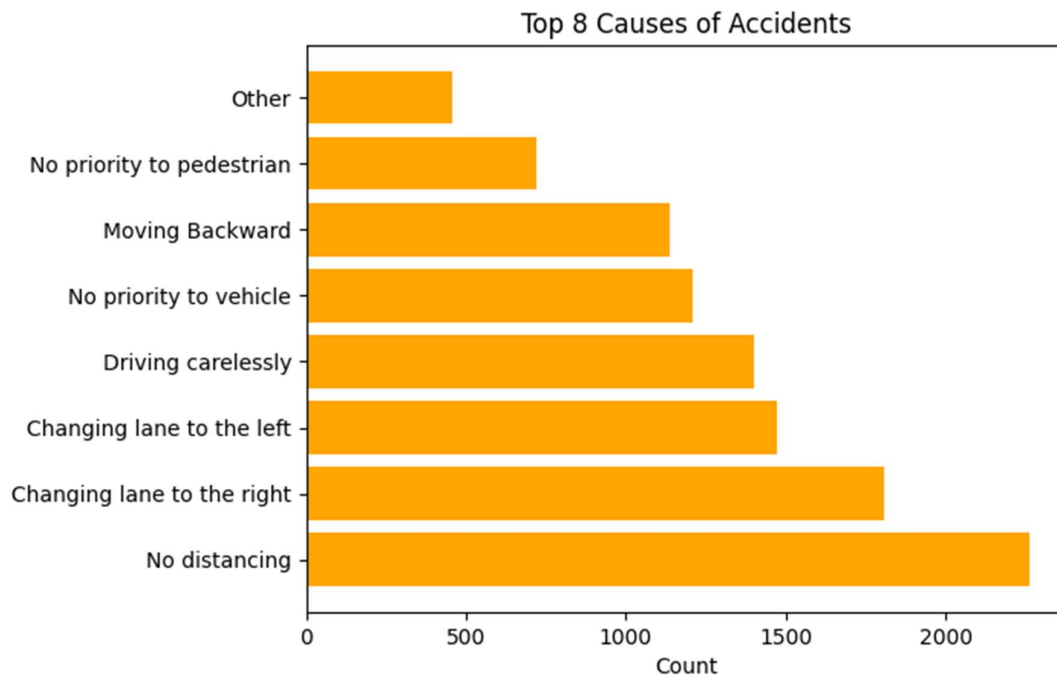


Accidents by Driver Age Band



Accidents by Day of Week

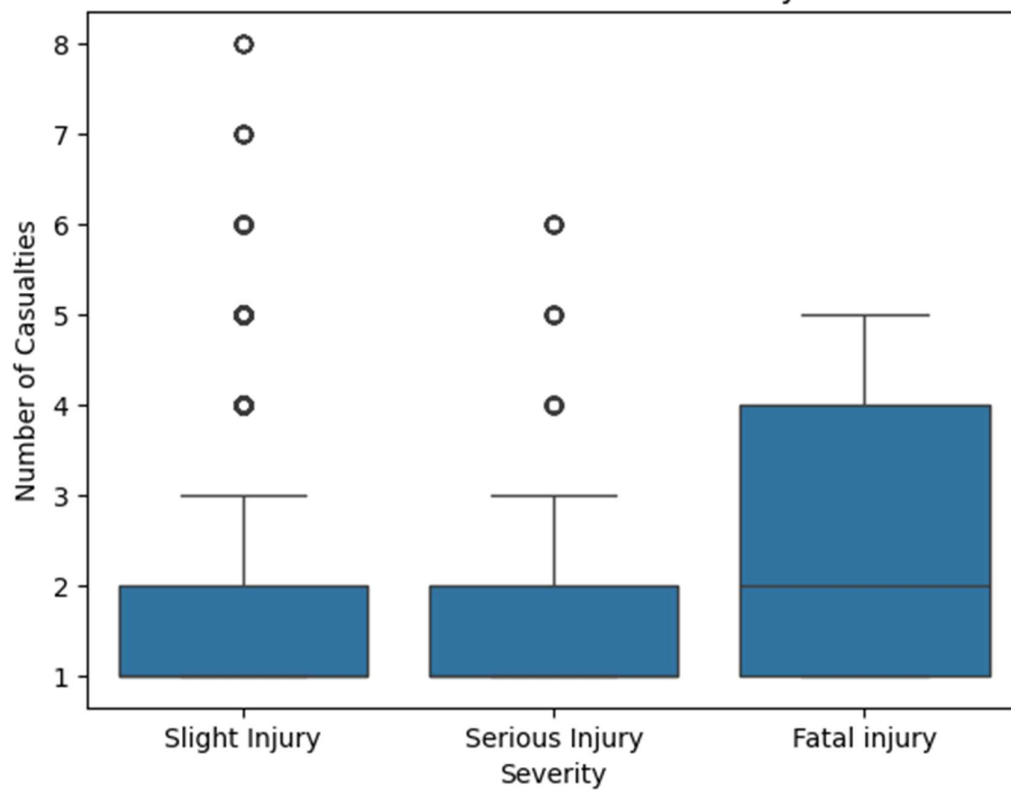




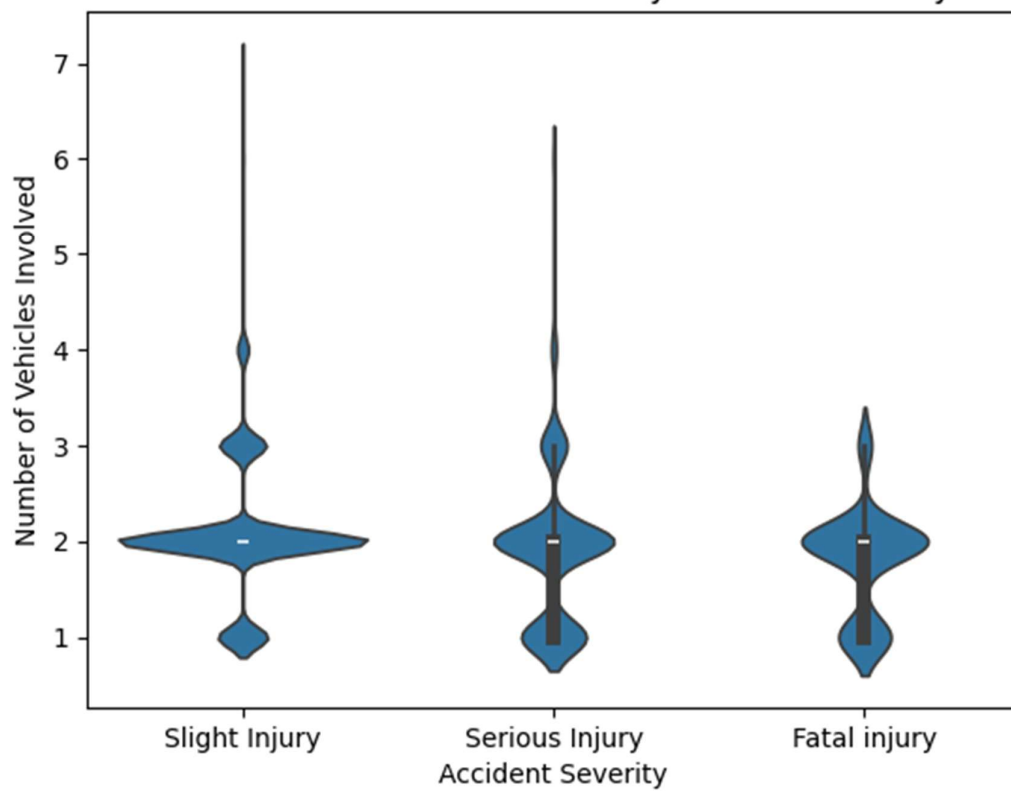
6.4 Bivariate Analysis

Bivariate analysis explores the relationship between two variables simultaneously. A boxplot comparing Number_of_casualties across different severity levels showed that fatal accidents, while rare, are associated with noticeably higher casualty counts. A violin plot was used to compare the distribution of Number_of_vehicles_involved across the three severity categories, showing that serious and fatal accidents tend to involve slightly more vehicles on average. A scatter plot of vehicles involved versus casualties confirmed a general positive trend: as more vehicles are involved in an accident, the total casualty count tends to rise. A grouped bar chart of severity by driver age band reinforced the finding that the 18–30 age group dominates across all severity categories.

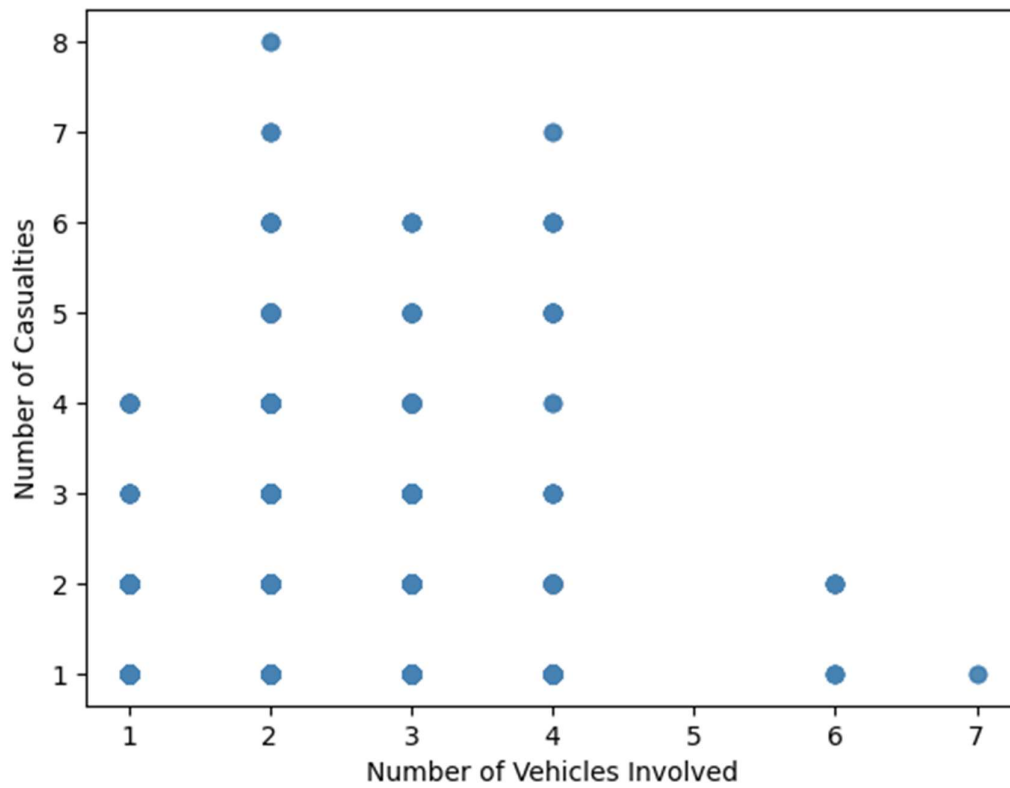
Casualties vs Accident Severity



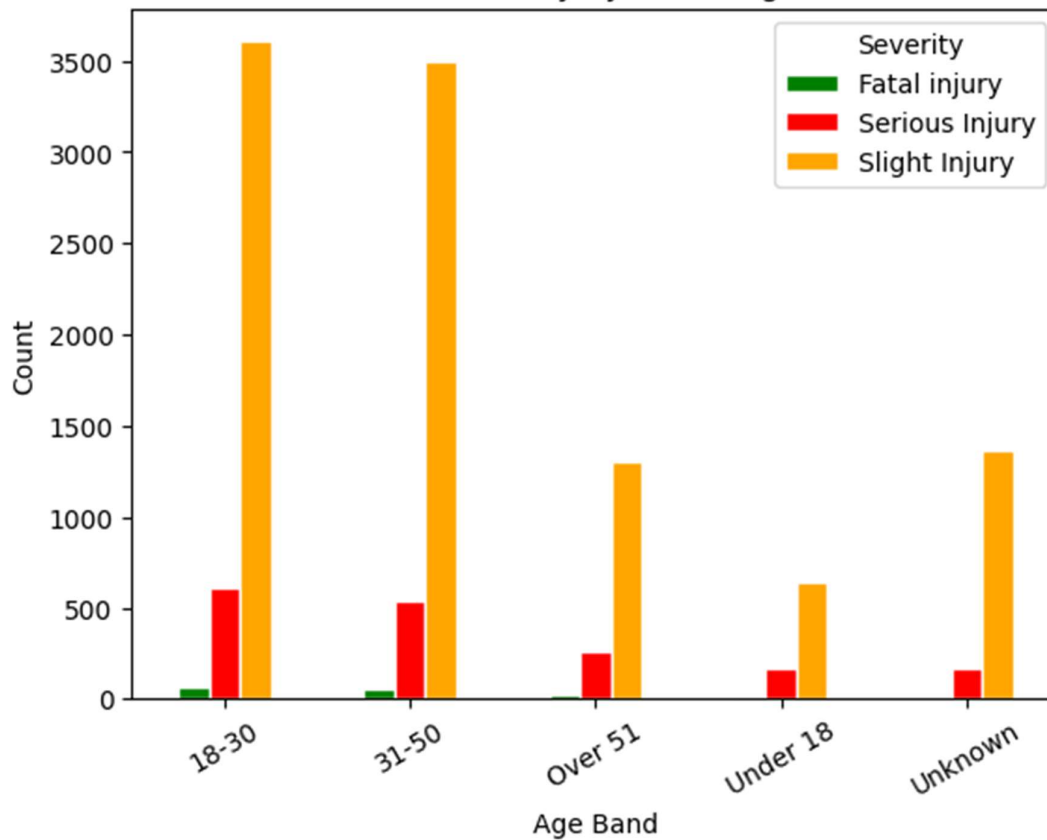
Vehicles Involved Distribution by Accident Severity



Vehicles Involved vs Number of Casualties

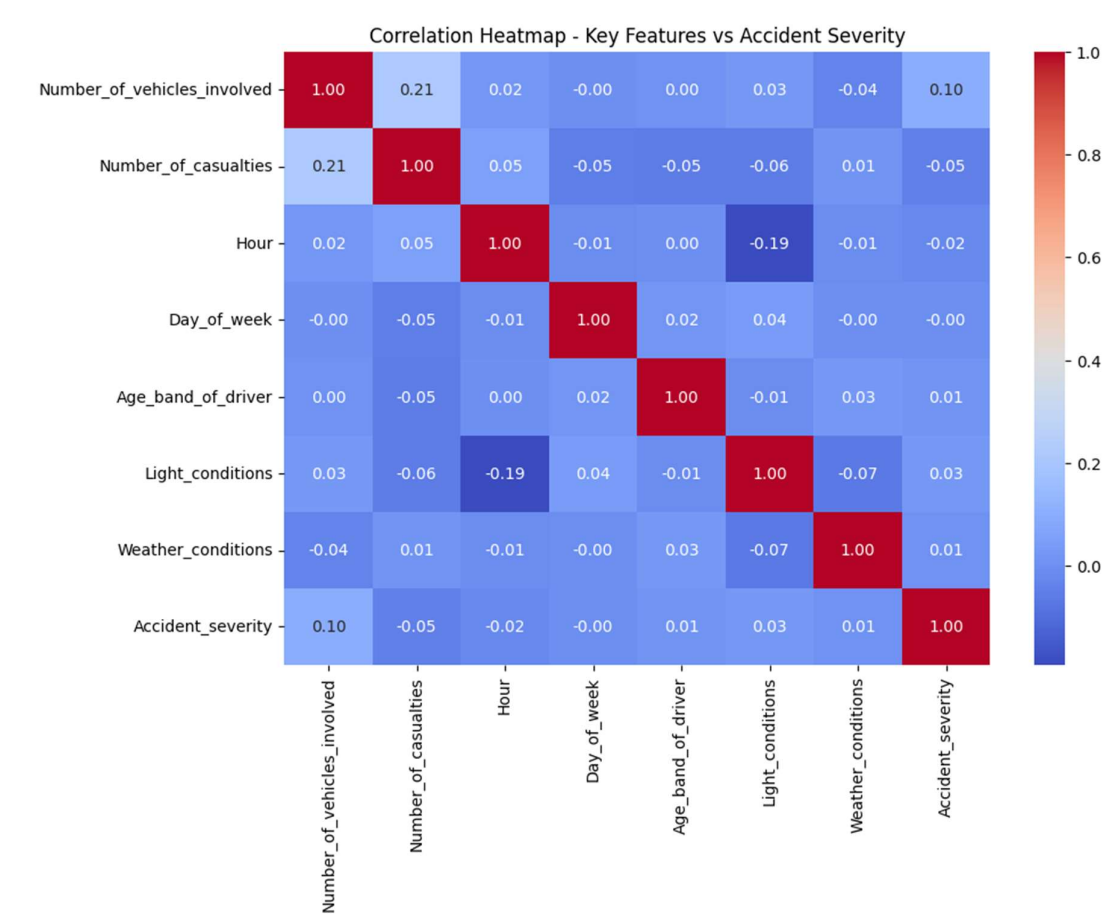


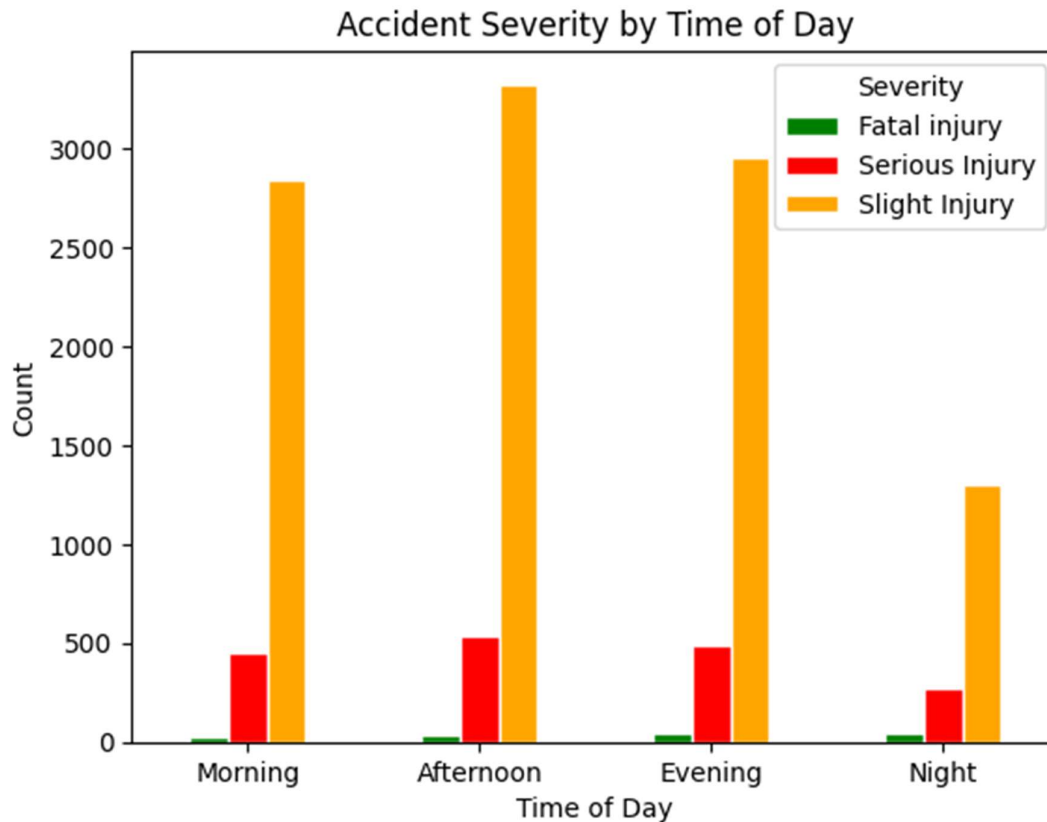
Accident Severity by Driver Age Band



6.5 Multivariate Analysis

Multivariate analysis examines interactions among multiple variables at once. A correlation heatmap was constructed using only the key encoded features — including Hour, Number_of_vehicles_involved, Number_of_casualties, Light_conditions, Weather_conditions, and Accident_severity — to identify which features are most strongly associated with the target. This confirmed that Number_of_casualties and Number_of_vehicles_involved carry the strongest signal. A grouped bar chart of severity by Time_of_Day showed that evening hours carry the highest absolute count of accidents across all severity levels, while night-time accidents, though fewer in number, show a proportionally higher rate of serious outcomes.





7. PREDICTIVE MODELING & COMPARISON STUDY

Not limiting the analysis to one algorithm, we can construct an arena of three entirely different machine learning models with their own philosophies when dealing with the prediction problem in question. In this way, we are not only finding the answer but proving that it is the best one possible.

The Three Models:

1. Logistic Regression Model (The Linear Boundary): The foundational block of any model, it uses a simple, linear boundary between classes created based on probability to output three categories of severity: slight injury, serious injury, or fatal injury.

2. Decision Tree Model (The Rule Engine): A rule engine for making predictions that creates an if-then decision tree from feature sets to classify the input data. It is very easy to understand yet vulnerable to overfitting with complex data.

3. Random Forest Classifier (The Ensemble): This is an advanced and ensemble form of the Decision Tree which trains hundreds of trees on random data samples and combines their votes together. Through the process of averaging individual errors, it is able to capture non-linear interactions between all 30+ variables much more effectively than any one tree.

All three algorithms were measured using Accuracy, MSE (Mean Squared Error), and MAE (Mean Absolute Error). In relation to forecasting road safety, minimising erroneous predictions, especially where there is underestimation of fatal and serious accidents, becomes paramount. As such, Random Forest performed exceptionally well with the least amount of errors and greatest level of accuracy, while the Decision Tree came in second and Logistic Regression failed to capture the non-linear nature of the classes within the data set.

8. INSIGHTS AND STORYTELLING

The data tells a clear story: young drivers (18–30) cause the most accidents, primarily due to avoidable behaviours which is failure to maintain safe following distance and unsafe overtaking. These two causes alone top the list, pointing directly to driver education as the highest-priority intervention.

Timing is a strong predictor. Accidents consistently peak during the 5pm to 7pm window (evening rush hour), and Friday records the most incidents. This pattern suggests targeted enforcement and awareness campaigns during Friday evenings would have the greatest impact.

The Number_of_casualties emerged as the strongest correlate of accident severity across all three correlation methods. Multi-vehicle collisions are disproportionately represented in serious and fatal outcomes, confirming that road safety measures reducing pile-up risk would directly lower fatality rates.

Despite the large dataset, fatal accidents represent only 1.3% of cases — yet this is not reassuring. It reveals that fatalities are largely preventable. The dominant causes are behavioural, not structural. Most serious incidents occur under normal road and weather conditions, meaning environmental factors are secondary to human error.

The machine learning models trained in this project further confirmed these patterns. Logistic Regression, Decision Tree, and Random Forest were each evaluated on accuracy, Mean Squared Error (MSE), and Mean Absolute Error (MAE). Random Forest, as expected from an ensemble method, delivered the best performance among the three. The model comparison charts clearly showed that simpler models like Logistic Regression struggle to capture the non-linear relationships inherent in this dataset, while tree-based methods leverage the rich categorical structure more effectively.

In summary, the data tells a story that is simultaneously predictable and urgent. Young drivers, distracted or overconfident, driving in the evening rush on Fridays, in multi-vehicle scenarios on busy roads, these are the conditions under which Ethiopia's most serious road accidents occur. This is not fate. It is a pattern, and patterns, once understood, can be interrupted.

9. CONCLUSION

The project has succeeded in converting the raw dataset into a useful format by carrying out various processes such as dealing with missing data, duplication, time feature engineering, and data encoding and scaling.

From the EDA conducted, specific patterns were identified; that is, younger drivers, nighttime periods, Fridays, and behavioral reasons (lack of distance, overtaking). These results are actionable since targeted policy interventions, law enforcement actions, and campaigns will help deal with these issues.

Out of the three Machine Learning models used in this case study, Random Forest algorithm performed very well. Also, a KNN classifier was introduced as another machine learning algorithm for predicting the outcome of interest. This shows that the prediction of accidents' severity is possible with some degree of accuracy.

In conclusion, the case study shows how valuable data science can be in terms of changing raw accidents information into actionable insights.